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United States Patent

12414722

Kind Code

B2

Date of Patent

September 16, 2025

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Portable cardiac monitoring device and associated cardiac monitoring system and network

Abstract

A device includes a main body having a length, a first end, a second end, and an axis along the length. A first electrode is associated with the first end of the main body. A suspension structure is associated with the second end of the main body. The suspension structure is configured for displacement in a direction of the axis of the main body, and includes a flexible isolation ring defining an opening, and a second electrode that is associated with the opening and mechanically coupled to the flexible isolation ring. The device also includes an electronics assembly with a first subassembly that is electrically coupled to the first electrode and secured relative to the main body to prevent displacement of the first subassembly in a direction of the axis of the main body, a second subassembly that is arranged to electrically couple with the second electrode. A flexible coupling between the first subassembly and the second subassembly enables displacement of the second subassembly relative to the first subassembly and in a direction of the axis of the main body.

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Appl. No.: 18/082500

Filed: December 15, 2022

Prior Publication Data

Document Identifier

Publication Date

US 20230200708 A1

Jun. 29, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **A61B5/332** (20210101); **A61B5/00** (20060101); **A61B5/26** (20210101); **A61B5/282** (20210101)

U.S. Cl.:

CPC **A61B5/332** (20210101); **A61B5/0006** (20130101); **A61B5/002** (20130101); **A61B5/26** (20210101); **A61B5/282** (20210101); **A61B5/6823** (20130101); **A61B5/6825** (20130101); **A61B5/7221** (20130101); A61B2560/0214 (20130101); A61B2560/0456 (20130101); A61B2562/0209 (20130101); A61B2562/166 (20130101); A61B2562/227 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/294,364, entitled “Portable Cardiac Monitoring Device and Associated Cardiac Monitoring System and Network” and filed on Dec. 28, 2021, which is expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to devices, systems, and networks for monitoring cardiac function, and more particularly, to a portable, handheld device for cardiac monitoring by a user outside a clinical setting and in association with a monitoring system and network.

BACKGROUND

(2) One of the greatest current challenges for medical systems worldwide is managing the growing number of patients with heart diseases. Regarding this important concern, the U.S. alone has millions of people with heart failure, and currently, over a half million new heart-failure patients are diagnosed each year. Heart failure is only one form of heart disease, but one of the most expensive ones for health care systems. The heart-failure challenge is growing globally with another sixty million, or so, patients outside of the U.S. About 50% of these other patients are in Asia, and the number in many developing countries is unknown due to the lack of diagnostic means and public information. Heart failure is a disease of the elderly, and it is growing rapidly in all societies with aging populations, and it is an epidemic burden for the health care system in every country.

(3) Key issues with the management of heart diseases arise both from the lack of simple, inexpensive, non-invasive ways to diagnose, evaluate, and monitor treatment for cardiac patients, and from the growing need to evaluate and monitor such patients as much as possible outside a clinical setting, e.g., hospital, physician's office, etc.

(4) Both acoustic cardiography and non-acoustic cardiography are low cost, non-invasive diagnostic technologies that enable fast assessment of a patient's cardiac function. These technologies provide nearly as much insight into heart function as ultrasound technology. A typical cardiography test is most often performed in the realm of acoustic cardiography and involves simultaneous acquisition of both electrocardiogram (ECG) data and heart-sound data, and analysis thereof. Non-acoustic cardiography involves acquisition and analysis of only ECG data.

(5) Known equipment employed for cardiography testing rely on relatively involved equipment models that require specific placements of electrodes on a patient's torso, and multi-wire connections to a data acquisition system. An example of such equipment used for non-acoustic cardiography is a Holter monitor, which uses electrodes attached to a patient's torso and a bulky recording device secured to the patient's waist to track cardiac activity for 24 to 72 hours.

(6) Other known equipment employed for cardiography testing are more portable in nature and do not require wired connections to a bulky recording device. An example of such equipment is the COALA™ Heart Monitoring System (www.coalalife.com). This system includes a handheld structure having three closely spaced electrodes that are held against the chest to sense ECG activity through narrow-field sensing vectors formed at the chest. An additional far field ECG recording is captured via electrode contact of fingertips from left and right hands. The handheld structure also includes a transducer for sensing heart sounds. Another example of portable cardiography testing equipment is the Eco DUO™ (www.ekohealth.com). This system includes a handheld structure having a two closely spaced electrodes that are held against the chest to sense ECG activity through a narrow-field sensing vector. The handheld structure also includes a

transducer for sensing heart sounds. The ECG sensing vectors enabled by these portable devices do not span the long axis of the heart, thereby limiting effective capture of electrical depolarization of the atria of the heart, e.g., p-waves. Accordingly, subsequent atrial fibrillation analysis by these devices will have diminished accuracy due to their inability to effectively correlate p-wave information.

(7) Another example of portable cardiography testing equipment that enables both acoustic cardiography and non-acoustic cardiography is disclosed in U.S. Pat. No. 9,451,896, which is hereby incorporated by reference. The portable device described in the '896 patent is in a form which accommodates both acoustic and non-acoustic cardiography by a patient outside of a clinical setting. The disclosed forms include a stylus-like configuration that resembles a pen or a pencil, and a finger-mountable ring configuration which is mountable on the outer end of a subject's finger. While this portable device enables ECG sensing through a far-field vector defined by a finger electrode and a chest electrode, each of these electrodes has a small service area and thus may not provide a quality ECG signal. Furthermore, the small electrodes and stylus configuration may be difficult for a user to hold in a stable manner for a duration of time needed to capture quality ECG signals and heart sound signals.

(8) It is therefore desirable to provide a portable cardiography device that provides a longer, far-field ECG vector that produces an ECG signal with less noise, and one in which the p-wave is more visible. The concepts disclosed below address these needs and others.

SUMMARY

(9) In one aspect, the disclosure relates to a portable monitoring device that includes a main body having a length, a first end, a second end, and an axis along the length. A first electrode is associated with the first end of the main body. A suspension structure is associated with the second end of the main body. The suspension structure is configured for displacement in a direction of the axis of the main body, and includes a flexible isolation ring defining an opening, and a second electrode that is associated with the opening and mechanically coupled to the flexible isolation ring. As used herein “displacement in a direction of the axis” may be expressed alternatively as movement up or down relative to the axis, or movement up or down along the axis. The device also includes an electronics assembly with a first subassembly that is electrically coupled to the first electrode and secured relative to the main body to prevent displacement of the first subassembly in a direction of the axis of the main body, a second subassembly that is arranged to electrically couple with the second electrode. A flexible coupling between the first subassembly and the second subassembly enables displacement of the second subassembly relative to the first subassembly and in a direction of the axis of the main body.

(10) In another aspect, the disclosure relates to a cardiac monitoring system that includes a portable monitoring device and dock-top case. The portable device has a main body with a charge-receiving coil. The dock-top case includes a cavity and a charge-transmitting coil that surrounding the cavity. The cavity is configured to receive a docking portion of the main body of the portable device such that the charge-receiving coil of the portable device aligns with the charge-transmitting coil of the case.

(11) In another aspect, the disclosure relates to a cardiac monitoring network that includes a portable monitoring device, a user device, and a network server. The portable device has a first electrode shaped to be placed in abutting contact with a palm of a user hand, and a second electrode shaped to be placed in abutting contact with a chest of the user while the first electrode in abutting contact with a palm of a user hand. The portable device also includes circuitry coupled to the first electrode and the second electrode that is configured to sense electrocardiogram (ECG) activity and communication circuitry configured to transmit a signal of the sensed ECG activity. The user device has communication circuitry configured to receive the signal of the sensed ECG activity from the portable device. The network server has a communication interface configured to communicate with the user device to receive the signal of the sensed ECG activity from the user

device.

(12) It is understood that other aspects of apparatuses and methods will become readily apparent to those skilled in the art from the following detailed description, wherein various aspects of apparatuses and methods are shown and described by way of illustration. As will be realized, these aspects may be implemented in other and different forms and its several details are capable of modification in various other respects. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of apparatuses and methods will now be presented in the detailed description by way of example, and not by way of limitation, with reference to the accompanying drawings, wherein:

(2) FIGS. 1A and 1B are perspective illustrations of a cardiac monitoring system that includes a portable device and a dock-top case for holding the device.

(3) FIG. 2 is a perspective, cross-section illustration of the cardiac monitoring system with the portable device placed in the dock-top case.

(4) FIG. 3A is a perspective, cross-section illustration of the portable device.

(5) FIG. 3B is a perspective, cross-section illustration of a portion of the portable device.

(6) FIGS. 4A and 4B are exploded illustrations of the portable device from different perspectives.

(7) FIG. 5 is a perspective, exploded illustration of the dock-top case of FIGS. 1A and 1B.

(8) FIG. 6 is a schematic diagram of the portable device.

(9) FIG. 7 is a block diagram of a cardiac monitoring network including a cardiac monitoring system, a user device, and a cloud based server/processor.

(10) FIG. 8 is a flowchart of use of the cardiac monitoring network of FIG. 7.

(11) FIGS. 9A and 9B are illustration of a portable device in use.

DETAILED DESCRIPTION

(12) Disclosed herein is a portable, handheld device that enables one or both of acoustic cardiography and non-acoustic cardiography using a single hand. The entirety of the device forms a bulb-like structure having a dome electrode at one end and a generally planar electrode at the opposite end. A grip ring, main housing, and flexible isolation ring extend between the dome electrode and the planar electrode. An interior space formed by the dome electrode, grip ring, and main housing houses electronics subassemblies that are coupled to electrodes for purposes of sensing electrical activity of the heart, and in some embodiments to an acoustic sensor for purposes of sensing acoustic activity of the heart. The electronics subassemblies may further include a processor for analyzing sensed electrical activity of the heart (referred to herein going forward as “ECG activity”), and if so configured, acoustic activity of the heart (referred to herein going forward as “heart-sound activity”) for diagnostic purposes. The electronics subassemblies may further include components that enable wireless communication with other devices. For example, the portable, handheld device may interface with a user device, e.g., a phone or other mobile device, which in turn interfaces with a network server/processor. Activity sensed by the portable device may be transmitted to the user device and provided to the network server/processor for analysis of and reporting on cardiac function.

(13) Due to the typical advanced age of the users of this device, the device has been designed to accommodate individuals with limited motor and dexterity due to physical limitations related to joint or neurological issues that affect controlled movements of the fingers and hands. To this end, the portable device disclosed herein is designed to accommodate various levels of hand dexterity limitations by providing a shape that can be easily held and manipulated with the large muscle

groups rather than relying on small muscle group control and strength. The geometry of the portable device does not require the user to hold and rotate the device in a specific orientation.

(14) With reference to FIGS. **9A** and **9B**, the shape of the portable device is designed to fit inside the natural relaxed state of the hand. The size is such that the portable device comfortably fits into as small as the five percentile and as large as the 95 percentile of hand sizes. The large dome of the portable device provides an electrode that is sized to create maximum surface contact of the palm of the hand, e.g., the right hand. The large contact surface of the dome electrode also allows for flexibility in the positioning of the portable device in the hand. The tapered transition and symmetry of the housing of the portable device allows the fingers to natural drape and contact the housing for ease of control and manipulation, and to comfortably rest on the unit to add additional stability. A rubberized grip of the portable device allows users to hold the device securely without slipping.

(15) Having generally described the geometry of the portable device and some of the benefits provided by the geometry, a more detailed description of the portable device within the contexts of a cardiac monitoring system and a cardiac monitoring network follow.

(16) With reference to FIGS. **1A** and **1B**, a cardiac monitoring system **100** includes a portable device **102** and a dock-top case **104** for holding the device. As described further below, the portable device **102** and the dock-top case **104** include respective charging coils that enable charging of a power supply of the device.

(17) With reference to FIGS. **2**, **3A**, **3B**, **4A**, **4B**, and **6**, the portable device **102** includes a main body **106**, a first electrode **108**, a suspension structure **110** with a second electrode **114**, and an electronics assembly **116** contained within an enclosure formed by the first electrode, the main body, and the suspension structure. A textured ring **148** may be located between the first electrode **108** and a portion of the main body **106**. The textured ring **148** provides enhanced gripping of the portable device **102**. Referring to FIG. **3A**, the main body **106** has a length, a first end **105**, a second end **107**, and an axis **136** along the length. In some embodiments, the main body has a tapered sidewall **150**. An illumination ring **160** may surround all or a portion of the main body **106** and may be just beneath the textured ring **148**. In one configuration, the illumination ring surrounds the entirety of the main body and is thus provides the ability for the user to see it at any angle while in use or in the dock-top case **104**.

(18) The first electrode **108** is associated with the first end **105** of the main body **106** and has a geometry that enables placement in the palm of a hand. In one configuration, the first electrode **108** has a non-planar hemispherical or dome shape. The first electrode **108** may be made of one or more metals, such as zinc, aluminum, magnesium, and/or nickel. The suspension structure **110** with the second electrode **114** is associated with the second end **107** of the main body **106** and is configured for displacement in a direction of the axis **136** of the main body. The second electrode **114** has a geometry that enables placement against a chest. In one configuration, the second electrode **114** has a planar disc shape. The second electrode **114** may be made of either carbon-filled plastic coated with Ag/AgCl or made of one or more metals, such as zinc, aluminum, magnesium, and/or nickel. The area of the exterior surface of the first electrode **108** is greater than the area of the exterior surface of the second electrode **114**, and in some embodiments is between 10 and 20 times greater. For example, in one configuration, the area of the exterior surface of the first electrode **108** is 5.8 square inches, while the patient contact area of the second electrode **114** is 0.3 square inches, which is similar in size as traditional ECG electrodes. Other size surface areas are contemplated, and the device disclosed herein is in no way limited to these example sizes.

(19) With reference to FIGS. **3A**, **3B**, **4A**, and **4B**, the suspension structure **110** includes a flexible isolation ring **112** defining an opening **113**. The second electrode **114** is associated with the opening and is mechanically coupled to the flexible isolation ring **112**. The flexible isolation ring **112** includes an undulating surface defined by concentric ribs **139** that forms a heart sound pickup when the second electrode **114** is pressed against the chest. The flexible isolation ring **112** is a solid

circular body that helps stabilize the second electrode **114** against the chest and provides a means to rest the portable device **102** against the chest without impeding the function of the heart sound pickup. The thickness of the second electrode **114** combined with the tension in the flexible isolation ring **112** ensure that the heart sound pickup makes positive contact with the body tissue for coupling heart sound vibration to the motion sensor **144**.

(20) Referring to FIG. 3B, regarding the mechanical coupling of the second electrode **114** and the flexible isolation ring **112**, the second electrode **114** includes a bottom plate **115** and an annular wall **117** extending from the upper surface of the bottom plate. The annular wall **117** includes a rim **119** at the top that extends outward from the annular wall. The bottom of the rim **119**, outer surface of the annular wall **117** and the upper surface of the bottom plate **115** define an annular groove of the second electrode **114**. This annular groove is sized to receive a portion **121** of the flexible isolation ring **112** and provides a mechanism through which the second electrode **114** is mechanically coupled to the flexible isolation ring in a way that enables easy removal and replacement. The interior of the annular wall **117** exposes a portion of the upper surface **142** of the bottom plate **115** and it is this surface against which the distal end **140** of the second printed circuit board **128** abuts.

(21) With reference to FIGS. 3A, 4A and 4B, the electronics assembly **116** includes a first subassembly **118**, a second subassembly **120**, and a flexible coupling **122** configured to electrically couple the first subassembly with the second subassembly. To this end, as shown in FIG. 3A, a first end of the flexible coupling **122** is electrically coupled to a coupling feature the first subassembly **118**, while a second end of the flexible coupling is electrically coupled to coupling of the second subassembly **120**. The flexible coupling **122** may have any one of a variety of configurations. For example, the flexible coupling **122** may be configured as a Z-bend flex wire structure, such as shown in FIGS. 3A, 4A and 4B. In other configurations, the flexible coupling may be a coiled wire structure.

(22) The first subassembly **118** is electrically coupled to the first electrode **108** and secured relative to the main body **106** to prevent displacement of the first subassembly in a direction of the axis **136** of the main body. In some embodiments the second subassembly **120** is fixedly electrically coupled with the second electrode **114**. In other embodiments, the second subassembly **120** is configured to electrically coupled with and decouple from the second electrode **114**. As shown in FIG. 3A, the flexible coupling **122** is located between the first subassembly **118** and the second subassembly **120** and is configured to enable up/down displacement of the second subassembly relative to the first subassembly and in a direction of the axis **136** of the main body **106**. The flexible coupling **122** is routed in a way to avoid introducing rubbing sounds that may interfere with the operation of the motion sensor **144**.

(23) Referring to FIGS. 2 and 4A, the first subassembly **118** of the electronics assembly **116** includes a first printed circuit board **124** and at least one first biasing mechanism **126**. The first biasing mechanism **126** is electrically coupled to the first printed circuit board **124** and extends from the printed circuit board into abutting contact with an electrically conductive region **132** of the first electrode **108**. The conductive region **132** may correspond to an interior surface of the first electrode **108**. The first biasing mechanism **126** may be a spring contact positioned relative to the interior surface **132** of the first electrode **108** so that the spring contact presses against the interior surface to form an electrical connection with the first electrode **108**. In some embodiments, the portable device **102** includes a pair of first biasing mechanism **126** for redundancy, each arranged to press against the interior surface **132** to form an electrical connection with the first electrode **108**.

(24) With reference to FIG. 6, in some embodiments the first subassembly **118** includes a controller **202**, ECG sensing circuitry **206**, heart sounds sensing circuitry **214**, communication circuitry **152**, a memory **204**, a power supply **154**, a charge-receiving coil **156**, and a haptic transducer **158**. The ECG sensing circuitry **206** is coupled to the first electrode **108** and the second electrode **114** and is

configured in accordance with standard, known ECG sensing technology to detect electrical activity of the heart through a sensing vector defined by the electrodes and to provide signals representing such activity, including for example cardiac cycles including atrial and ventricular depolarizations and repolarizations. The heart sounds sensing circuitry **214** is coupled to the motion sensor **144** and is configured in accordance with standard, known heart sounds sensing technology to detect heart sounds through the acoustic sensor and to provide signals presenting heart sounds, including for example cardiac cycles including S1 through S4 heart sounds.

(25) Continuing with reference to FIG. **6**, the controller **202** is configured to receive the signals representing sensed ECG activity from the ECG sensing circuitry **206**. In some embodiments, the controller **202** is further configured to create a communications data packet that includes the signals representing sensed ECG activity and to transmit the data packet through the communication circuitry **152**. In some embodiments, the controller **202** is further configured to create a memory data packet that includes the signals representing sensed ECG activity and to store the data packet in memory **204** for later retrieval therefrom and transmission through the communication circuitry **152**.

(26) Likewise, the controller **202** is configured to receive the sensed heart sounds activity from the heart sounds sensing circuitry **214**. In some embodiments, the controller **202** is further configured to create a communications data packet that includes the signals representing the sensed heart sounds activity and to transmit a communications data packet through the communication circuitry **152**. In some embodiments, the controller **202** is further configured to create a memory data packet that includes the signals representing sensed heart sounds activity and to store the data packet in memory **204** for later retrieval therefrom and transmission through the communication circuitry **152**.

(27) The communication circuitry **152** includes a receiver/transmitter and antenna, and may be a Bluetooth Low Energy Radio or other wireless communications technology. The first subassembly **118** may also include a motion detector **208** that detects removal of the portable device **102** from the dock-top case **104** and activates the power supply **154** to thereby turn on the device. The power supply **154** may be a capacitor. The charge-receiving coil **156** is configured to receive charging energy from a charge-transmitting coil of the dock-top case **104**.

(28) Referring to FIGS. **2**, **3A**, and **3B**, the second subassembly **120** of the electronics assembly **116** includes a second printed circuit board **128** and, in some embodiments, a second biasing mechanism **130**. The second biasing mechanism **130** is electrically coupled to the second printed circuit board **128** and extends from the second printed circuit board to a location relative to an electrically conductive region **134** of the second electrode **114**. The second biasing mechanism **130** is located relative to the electrically conductive region **134** of the second electrode **114** such that a displacement of the suspension structure **110** along the axis **136** of the main body **106** enables the second electrode **114** to transition between an engaged state and a disengaged state. In an engaged state, the second biasing mechanism **130** is electrically coupled with the electrically conductive region **134** of the second electrode **114**. In a disengaged state, the second biasing mechanism **130** is electrically decoupled from the electrically conductive region **134** of the second electrode. In other embodiments, the second subassembly **120** does not include a second biasing mechanism **130** and electrical coupling between the second printed circuit board **128** and the second electrode **114** is provided by soldering the second printed circuit board to the second electrode.

(29) With reference to FIG. **3B**, the suspension structure **110** is configured to transition the second electrode **114** from the disengaged state to the engaged state responsive to an application of a sufficient force to an exterior surface **138** of the suspension structure **110**. The force may be applied to one or more ribs **139** of the flexible isolation ring **112** and the bottom plate **115** of the second electrode **114**. The force may result, for example, from the pressing of the bottom plate **115** of the second electrode **114** and the suspension structure **110** against the chest. Conversely, the suspension structure **110** is configured to transition the second electrode **114** from the engaged state to the

disengaged state responsive to a removal of the sufficient force from the bottom plate **115** of the second electrode **114** and the exterior surface **138** of the suspension structure. The removal of such force may result, for example, from the removing of the suspension structure **110** from the chest. (30) With reference to FIGS. **3A** and **3B**, the second printed circuit board **128** has a distal end **140** in abutting contact with an upper surface **142** of the second electrode **114**. The second subassembly **120** of the electronics assembly **116** also includes a motion sensor **144** associated with the second printed circuit board **128**. The motion sensor **144** may be, for example, a low-noise MEMS accelerometer. Displacement of the suspension structure **110** in the direction of the axis **136** of the main body **106** results in displacement of the second electrode **114** and corresponding displacement of the second printed circuit board **128** and the motion sensor **144**. The second printed circuit board **128**, the second electrode **114**, and the flexible isolation ring **112** form a movable portion of the portable device **102** that has a center of mass in alignment or near alignment with a center line **146** extending through the center of the suspension structure **110** in a direction of the axis **136** of the main body **106**. This movable portion may be referred to as a transducer system. Keeping the center of mass of the transducer system in alignment or near alignment with the center line **146** avoids rotational movement of the second printed circuit board **128** in response to vibration forces. The mass of the movable portion (e.g., the second printed circuit board **128**, the second electrode **114**, and the flexible isolation ring **112**) combined with the elasticity of the body tissue against which the second electrode is pressed during use, form a low-pass filter limiting the usable signal bandwidth of the transducer system. By keeping the mass of the movable portion low, a desired bandwidth is accomplished.

(31) Thus, disclosed herein is a portable device **102** that provides for far-field sensing of ECG activity through a vector formed by a first electrode site at the hand and a second electrode site at the chest. In comparison to near-field vectors formed between two chest electrodes, the vector of the disclosed portable device **102** produces signals of ECG activity with less noise, and ones in which atrial events, e.g., p-waves, are more visible. The dome shape of the first electrode provides a large surface area that enables the capture of quality ECG activity without the need for gel and a shape that provides an ergonomic fit and grip with hand.

(32) With reference to FIGS. **2** and **5**, the dock-top case **104** includes a housing **170**, a charging coil assembly **172**, a dock charging printed circuit board **174**, a case bottom **176**, and a rubber base pad **178**. The charging coil assembly **172** includes a cylindrical structure **180** having an upper annular flange **184** that supports a charge-transmitting coil **182**. The charge-transmitting coil **182** is electrically coupled to the dock charging printed circuit board **174** through conductors **186**. As shown in FIG. **2**, when assembled, the housing **170** and the cylindrical structure **180** of the charging coil assembly **172** form a cavity **188** configured to receive a docking portion of a portable device **102**. Respective dimensions of the cavity **188** and a docking portion of a portable device **102** are such that the charge-transmitting coil **182** of the dock-top case **104** and the charge-receiving coil **156** of the portable device are vertically aligned to facilitate charging.

(33) More specifically, the height of the cavity **188** from the top rim **190** of the dock-top case **104** to the bottom of the bottom **192** of the cylindrical structure **180**, and the diameter of the opening at the top rim are sized relative to the height of the portable device **102** between the bottom of the second electrode **114** and the junction **194** of the tapered sidewall **150** and the illumination ring **160** and the diameter of the portable device **102** at the junction, such that when the portable device is placed in the cavity **188** of the dock-top case **104**, the portion of the portable device at the junction **194** is in abutting contact with a tapered region **196** of the dock-top case, and the charge-receiving coil **156** of the portable device is vertically aligned with the charge-transmitting coil **182** of the dock-top case, and the bottom of the second electrode **114** of the portable device is suspended above the bottom **192** of the cylindrical structure **180**.

(34) Thus, disclosed herein a cardiac monitoring system **100** that includes a portable device **102** and dock-top case **104**. The portable device **102** has a main body **106** with a charge-receiving coil

156. The dock-top case **104** includes a cavity **188** and a charge-transmitting coil **182** that surrounding the cavity. The cavity **188** is configured to receive a docking portion of the main body **106** of the portable device **102** such that the charge-receiving coil **156** of the portable device aligns with the charge-transmitting coil **182** of the case.

(35) FIG. 7 is a block diagram of a cardiac monitoring network **700** that includes a cardiac monitoring system **100**, a user device **210**, and a cloud based server/processor **212**. The cardiac monitoring system **100** includes a portable device **102** and a dock-top case **104**. An example application of the cardiac monitoring system **100** is described below with reference to FIG. 8.

(36) With reference to FIG. 8, an example application of the cardiac monitoring system of FIGS. 1-6 within a cardiac monitoring network **700** is now described.

(37) At block **802**, an application is started on a user device **210**. The user device **210** may be a phone, tablet, laptop, or desktop computer.

(38) At block **804**, a portable device **102** is connected to the user device **210**. To this end, a user may remove the portable device **102** from the dock-top case **104**. Upon removal from the dock-top case **104**, the portable device **102** automatically turns on and connects to the user device **210** via the communication circuitry **152**.

(39) At block **806**, the user device **210** is connected to a server/processor **212** of the cardiac monitoring network **700**. The connection between the user device **210** and the server/processor **212** may be made using standard communication technologies, including for example, wireless cellular technology or wired Internet technology.

(40) At block **808**, test signals corresponding to sensed ECG activity and sensed heart sounds activity are captured. To this end, and with particular reference to FIGS. 9A and 9B, the portable device **102** is held with the right hand and pressed lightly against the chest in the V3 or V4 precordial lead location for approximately 10 seconds. During this time, the portable device **102** captures separate ECG activity and heart sounds activity signals and transmits each of these signals to the user device **210** in real-time. The user device **210**, in turn, transmits the respective ECG activity and heart sounds activity signals to the server/processor **212** of the cardiac monitoring network **700** in real-time.

(41) At block **810**, a determination is made regarding proper locating of the second electrode **114** on the chest and the quality of the ECG activity signal and the heart sounds activity signal. In some embodiments, the server/processor **212** may analyze each of the ECG test signal and the heart sounds test signal to determine if the electrodes **108**, **114** and the suspension structure **110** of the portable device **102** are being held in positions that establish a proper ECG sensing vector and/or a proper audible capture of heart sounds. This is accomplished by comparing morphologic characteristics of the test signals to “fingerprint” signals previously captured during setup and patient training.

(42) If it is determined that the quality of one or more of the test signal of ECG activity and the test signal of heart sounds activity is not acceptable, then the process proceeds to block **812** where instructions are received at the user device **210** from the server/processor **212** to the user device **210** regarding the positions of the electrodes **108**, **114** and repositioning of the portable device **102**. For example, one or more of an audible, visual, or tactile (haptic) feedback may be provided to the user, prompting them to adjust the position of the suspension structure **110** relative to the chest, and with it the location of the second electrode **114**. Audible and visual feedback may be provided through the user device **210**. Visual feedback may be provided by the illumination ring **160** of the portable device. Tactile feedback may be provided by a vibration mechanism or haptic transducer **158** of the portable device **102**. The process then returns to block **808** where another set of test signals of ECG activity and heart sounds activity are captured after the position of the suspension structure **110** and the second electrode **114** are adjusted.

(43) Returning to block **810**, if it is determined that the quality of each of the ECG signal and the heart sounds signal is acceptable, then the process proceeds to block **814** where analysis signals of

ECG activity and heart sounds activity are captured. To this end, the portable device **102** is maintained at its current position and continues to be held with the right hand and pressed lightly against the chest in the V3 or V4 precordial lead location for a minute or so. During this time, the portable device **102** captures signals of ECG activity and heart sounds activity and transmits them to the user device **210** in real-time. Alternatively, the signals of ECG activity and heart sounds activity may be stored in the memory **204** of the portable device **102** for subsequent transmission to the user device **210**.

(44) In some embodiments, upon receipt of the analysis signals of ECG activity and heart sounds activity, the user device **210** transmits the signals to the server/processor **212** of the cardiac monitoring network **700** in real-time. Alternatively, the signals of ECG activity and heart sounds activity may be stored in a memory of the user device **210** for subsequent transmission to the server/processor **212** of the cardiac monitoring network **700**. In either case, upon receipt of qualified signals of ECG activity and heart sounds activity, the server/processor **212** analyzes the signals and transmits result to the user device **210**.

(45) Continuing with block **814**, after capture and transmission of the analysis signals of ECG activity and heart sounds activity, the portable device **102** may be placed in the dock-top case **104** for storage and recharging.

(46) Thus, disclosed herein is a cardiac monitoring network **700** that includes a portable monitoring device **102**, a user device **210**, and a network server **212**. The portable device **102** has a first electrode **108** shaped to be placed in abutting contact with a palm of a user hand, and a second electrode **114** shaped to be placed in abutting contact with a chest of the user while the first electrode in abutting contact with a palm of a user hand. The portable device **102** also includes circuitry coupled to the first electrode **108** and the second electrode **114** that is configured to sense electrocardiogram (ECG) activity and communication circuitry **152** configured to transmit a signal of the sensed ECG activity. The user device **210** has communication circuitry configured to receive the signal of the sensed ECG activity from the portable device **102**. The network server **212** has a communication interface configured to communicate with the user device **210** to receive the signal of the sensed ECG activity from the user device.

(47) The various aspects of this disclosure are provided to enable one of ordinary skill in the art to practice the present invention. Various modifications to exemplary embodiments presented throughout this disclosure will be readily apparent to those skilled in the art. Thus, the claims are not intended to be limited to the various aspects of this disclosure, but are to be accorded the full scope consistent with the language of the claims. All structural and functional equivalents to the various components of the exemplary embodiments described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

Claims

1. A device comprising: a main body having a length, a first end, a second end, and an axis along the length; a first electrode associated with the first end of the main body; a suspension structure associated with the second end of the main body, the suspension structure configured for displacement in a direction of the axis of the main body, the suspension structure including: a flexible isolation ring defining an opening, and a second electrode associated with the opening and mechanically coupled to the flexible isolation ring; and an electronics assembly comprising: a first subassembly electrically coupled to the first electrode and secured relative to the main body to prevent displacement of the first subassembly in a direction of the axis of the main body, a second

subassembly arranged to electrically couple with the second electrode, and a flexible coupling between the first subassembly and the second subassembly configured to enable displacement of the second subassembly relative to the first subassembly and in a direction of the axis of the main body.

2. The device of claim 1, wherein the first subassembly comprises: a first printed circuit board; and a first biasing mechanism electrically coupled to the first printed circuit board and extending from the first printed circuit board into abutting contact with an electrically conductive region of an interior surface of the first electrode.

3. The device of claim 1, wherein the second subassembly comprises: a second printed circuit board; and a second biasing mechanism electrically coupled to the second printed circuit board and extending from the second printed circuit board to a location relative to an electrically conductive region of the second electrode.

4. The device of claim 3, wherein the second printed circuit board has a distal end in abutting contact with an inner surface of the second electrode, and the second subassembly further comprises: a motion sensor associated with the second printed circuit board; wherein displacement of the suspension structure in the direction of the axis of the main body comprises displacement of the second electrode and corresponding displacement of the second printed circuit board and the motion sensor.

5. The device of claim 4, wherein the second printed circuit board has a center of mass aligned with a center line extending through the center of the suspension structure in a direction of the axis of the main body.

6. The device of claim 1, wherein the flexible coupling of the electronics assembly is configured to electrically couple the first subassembly with the second subassembly.

7. The device of claim 6, wherein the flexible coupling comprises one of a coiled wire structure or a Z-bend flex wire structure.

8. The device of claim 1, wherein the first electrode has an exterior surface area in a range of 10 to 20 times greater than an exterior surface area of the second electrode.

9. The device of claim 1, wherein the first electrode is characterized by a non-planar geometric shape.

10. The device of claim 9, wherein the non-planar geometric shape is a dome.

11. The device of claim 1, wherein the second electrode is planar.

12. The device of claim 1, further comprising a textured ring between the first electrode and a portion of the main body.

13. The device of claim 1, wherein the main body has a tapered sidewall.
